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Method for producing transmon qubit and lithium niobate resonator on the same substrate

Abstract

A fabrication method and associated apparatus is disclosed where an electromechanical resonator made out of lithium niobate is fabricated on the same substrate as a Josephson Junction-based transmon qubit. The starting material may be a high resistivity silicon wafer with a thin layer of lithium niobate (LiNbO_3). The fabrication method may include removing lithium niobate selectively from the substrate to preserve the quality of the substrate. The selective removal maintains defect free qualities of the silicon surface, thus enabling the fabrication of high performance Josephson Junction-based transmon qubit on the surface.

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Background/Summary

BACKGROUND

(1) Quantum computing utilizes the laws of quantum physics to process information. Quantum mechanics is a theory that describes the behavior of reality at the fundamental level. It is currently the only physical theory that is capable of consistently predicting the behavior of microscopic quantum objects like photons, molecules, atoms, and electrons.

(2) A quantum computer is a device that utilizes quantum mechanics to allow one to write, store, process and read out information encoded in quantum states, i.e., the states of quantum objects. A quantum object is a physical object that behaves according to the laws of quantum physics. The state of a physical object is a mathematical description of the object at a given time.

(3) In quantum mechanics, the state of a two-level quantum system is a quantum bit, or qubit, and is a list of two complex numbers whose squares sum up to one. Each of the two numbers is called an amplitude, and their squared absolute values are the probabilities that a measurement of the qubit results in zero or one, respectively. A fundamental and counterintuitive difference between a probabilistic bit (e.g., a zero or one bit encoded in a system described by classical physics, like a coin) and the qubit is that a probabilistic bit represents a lack of information about a two-level classical system, while a qubit contains maximal information about a two-level quantum system.

(4) Quantum computers are based on qubits, which may experience the phenomena of “superposition” and “entanglement.” Superposition allows a quantum system to be in multiple states at the same time. For example, whereas a classical computer is based on bits that are either zero or one, a qubit may be both zero and one at the same time, with different probabilities assigned to zero and one. Entanglement is a strong correlation between quantum systems, such that the quantum systems are inextricably linked even if separated by great distances.

(5) A quantum algorithm is a reversible transformation acting on qubits in a desired and controlled way, followed by a measurement on one or multiple qubits. For example, if a system has two qubits, a transformation may modify four numbers; with three qubits this becomes eight numbers, and so on. As such, a quantum algorithm acts on a list of numbers that is exponentially large as dictated by the number of qubits. To implement a transform, the transform may be decomposed into small operations acting on a single qubit, or a set of qubits, as an example. Such small operations are called quantum gates and a specific arrangement of the gates implements a quantum algorithm.

(6) There are different types of two-level quantum objects that may be used in quantum computers, each having different advantages and disadvantages. For example, some quantum computers may include qubits encoded in superconductors, trapped ions, electron spins, photons, etc. Each may experience different levels errors and decoherence. Also, some may be more useful for implementing particular types of quantum algorithms, while others may be more useful for implementing other types of quantum algorithms. Also, costs, run-times, error rates, availability, etc. may vary across quantum computing technologies. One approach that is actively being researched uses nonlinear superconducting circuits coupled to resonators. The resonators are used

as memory elements to store qubits and can be of any type, such as a superconducting cavities or mechanical resonators.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 depicts a cross-sectional representation of an embodiment of a substrate for a quantum computing device.
- (2) FIG. 2. depicts a cross-sectional representation of a mask formed over a lithium niobate layer, according to some embodiments
- (3) FIG. 3 depicts a cross-sectional representation of a proton exchange treatment, according to some embodiments.
- (4) FIG. 4 depicts a cross-sectional representation of a substrate after a wet etch process, according to some embodiments.
- (5) FIG. 5 depicts a cross-sectional representation of a substrate with a lithium niobate structure mask, according to some embodiments.
- (6) FIG. 6 depicts a cross-sectional representation of a substrate with a lithium niobate structure mask and a silicon substrate mask, according to some embodiments.
- (7) FIG. 7 depicts a cross-sectional representation of a substrate with a lithium niobate structure mask and a silicon substrate mask being subjected to a dry etch process, according to some embodiments.
- (8) FIG. 8 depicts a cross-sectional representation of a substrate with a lithium niobate structure formed by a dry etch process and after mask removal, according to some embodiments.
- (9) FIG. 9 depicts a cross-sectional representation of a substrate with a superconducting electrode layer formed on the substrate, according to some embodiments.
- (10) FIG. 10 depicts a cross-sectional representation of a substrate with a nonlinear element formed on the substrate, according to some embodiments.
- (11) FIG. 11 depicts a cross-sectional representation of a substrate with a bandage layer formed on the substrate, according to some embodiments.
- (12) FIG. 12 depicts a cross-sectional representation of a substrate with an acoustic resonator and a nonlinear element formed on the substrate, according to some embodiments.
- (13) FIG. 13 depicts a cross-sectional representation of a substrate with masks formed on a lithium niobate layer, according to some embodiments.
- (14) FIG. 14 depicts a cross-sectional representation of the substrate from FIG. 13 being dry etched, according to some embodiments.
- (15) FIG. 15 depicts a cross-sectional representation of the substrate from FIG. 13 after dry etching and mask removal, according to some embodiments.
- (16) FIG. 16 depicts a cross-sectional representation of the substrate from FIG. 15 with a mask formed, according to some embodiments.
- (17) FIG. 17 depicts a cross-sectional representation of the substrate from FIG. 16 during proton exchange treatment, according to some embodiments.
- (18) FIG. 18 depicts a cross-sectional representation of the substrate from FIG. 16 after proton exchange, according to some embodiments.
- (19) FIG. 19 depicts a cross-sectional representation of the substrate from FIG. 18 after removal of proton exchange material and the mask, according to some embodiments.
- (20) FIG. 20 depicts a cross-sectional representation of a substrate with an acoustic resonator and a nonlinear element formed on the substrate of FIG. 19, according to some embodiments
- (21) FIG. 21 is a flow diagram illustrating a method of fabrication of a device, according to some embodiments.

(22) FIG. 22 is a block diagram illustrating an example computing device that may be used in at least some embodiments.

(23) While embodiments are described herein by way of example for several embodiments and illustrative drawings, those skilled in the art will recognize that embodiments are not limited to the embodiments or drawings described. It should be understood, that the drawings and detailed description thereto are not intended to limit embodiments to the particular form disclosed, but on the contrary, the intention is to cover all modifications, equivalents and alternatives falling within the spirit and scope as defined by the appended claims. The headings used herein are for organizational purposes only and are not meant to be used to limit the scope of the description or the claims. As used throughout this application, the word “may” is used in a permissive sense (i.e., meaning having the potential to), rather than the mandatory sense (i.e., meaning must). Similarly, the words “include,” “including,” and “includes” mean including, but not limited to. When used in the claims, the term “or” is used as an inclusive or and not as an exclusive or. For example, the phrase “at least one of x, y, or z” means any one of x, y, and z, as well as any combination thereof.

DETAILED DESCRIPTION

(24) The present disclosure relates to fabrication methods for quantum computing devices that utilize electromechanical resonators.

(25) One of the most popular implementations of quantum computing are superconducting circuits. They are often comprised of a Josephson Junction shunted by a capacitor, a configuration typically referred to as a transmon qubit. Certain quantum processor architectures utilize, as a basic building block, a linear resonator strongly coupled to a transmon or some other kind of nonlinear superconducting circuit. Electromechanical resonators have multiple advantages over purely electrical ones such as small size and potentially superior quality factors. However, it has been challenging to integrate electromechanical resonator and a transmon within the same fabrication process flow, while preserving transmon performance and strong coupling between the elements.

(26) Transmon performance, which is typically measured by coherence times $T_{\text{sub.1}}$ and $T_{\text{sub.2}}$, is very sensitive to substrate and material purity. Therefore, a very specific set of procedures, involving substrate cleaning and special surface treatments, often needs to be performed during device fabrication. One way to provide strong coupling between mechanical motion and electrical charge is via the piezoelectric effect. This way, however, requires the processing of piezoelectric material on the same substrate where the transmon is built. Processing of piezoelectric material on the same substrate where the transmon is built (e.g., on the same silicon substrate) may be difficult to implement while maintaining suitable operating properties for a quantum computing device.

(27) In various implementations, lithium niobate can be grown as high purity crystal boules using the Czochralski method. Then, a thin layer of pure lithium niobate is bonded to a high resistivity silicon wafer using, for example, a smart-cut technique. High resistivity silicon wafers with low dopant and low interstitial oxygen concentrations have been shown to provide a great substrate for transmon qubits with $T_{\text{sub.1}}$ times of order of 100 microseconds (μs). However, if the pristine silicon surface is subjected to any physical ion bombardment during processing, such as during over-etch while patterning the lithium niobate film, the performance is degraded. A commonly known method to pattern lithium niobate involves dry etching with low selectivity to silicon. This process, however, subjects the silicon substrate surface to excessive ion bombardment, thereby degrading transmon performance.

(28) The present disclosure provides a fabrication method where lithium niobate is selectively patterned (lithium niobate being one of the piezoelectric materials with the largest electromechanical coupling) to preserve the quality of the substrate (e.g., high resistivity silicon). For instance, a proton exchange treatment process, as described herein, may be implemented for selectively patterning of a layer of lithium niobate on a silicon substrate. Quantum computing devices (such as nonlinear superconducting elements) made by the disclosed method demonstrate high $T_{\text{sub.1}}$ and $T_{\text{sub.2}}$ for a transmon fabricated on a substrate where lithium niobate was

selectively etched.

(29) FIGS. **1-12** depict cross-sectional representations of various steps in a method for fabricating a quantum computing device on a silicon substrate. FIG. **12** depicts an example of a quantum computing device fabricated according to the disclosed embodiments. FIG. **1** depicts a cross-sectional representation of an embodiment of a substrate **100** for a quantum computing device. In the illustrated embodiment, substrate **100** includes silicon substrate **102** with lithium niobate layer **104** coupled to (e.g., bonded to) the silicon substrate. Silicon substrate **102** may be, for example, a bulk silicon substrate with high resistivity. In various embodiments, silicon substrate **102** has a surface roughness on the order of at most about 1 nm.

(30) In various embodiments, lithium niobate layer **104** is a layer of single crystal lithium niobate. For instance, lithium niobate layer **104** may be a X-cut lithium niobate layer. Lithium niobate layer **104** may be thin compared to the thickness of silicon substrate **102**. For example, in one embodiment, lithium niobate layer **104** has a thickness on the order of 250 nm. Other thicknesses may, however, also be implemented depending on the desired operating properties of a quantum computing device formed on substrate **100**. For example, the thickness of lithium niobate layer **104** may vary between about 100 nm and about 500 nm or other ranges therein.

(31) Typically, substrate **100** (e.g., silicon substrate **102** with lithium niobate layer **104** bonded to the silicon substrate) may be prefabricated and procured from vendors known in the surface acoustic wave industry. For instance, a substrate **100** may be obtained from NGK Insulators, LTD (Japan). In some embodiments, a lithium niobate layer **104** in substrate **100** may be thicker than desired on a prefabricated substrate. In such embodiments, lithium niobate layer **104** may be dry etched to thin the layer to a desired thickness.

(32) While embodiments disclosed herein implement lithium niobate as the material for generating an electromechanical (acoustic) resonator, other materials may also be contemplated without deviating from the scope of the fabrication processes and devices described herein. For instance, in certain embodiments, lithium niobate is a protonatable piezoelectric material. As used herein, a “protonatable piezoelectric material” is a piezoelectric material that is proton exchangeable. Thus, protonatable piezoelectric material may be used interchangeably with lithium niobate in the fabrication processes and devices described herein. Examples of protonatable piezoelectric materials include, but are not limited to, lithium-based salts (such as lithium niobate or lithium tantalate) or other lithium-based materials that are protonatable. In certain embodiments, a protonatable piezoelectric material is a single crystal material. For example, the protonatable piezoelectric material may be a single crystal lithium-based salt. Embodiments with non-single crystal protonatable piezoelectric material may also be contemplated.

(33) In certain embodiments, a mask is placed over lithium niobate layer **104** to define a pattern of one or more islands of lithium niobate on silicon substrate **102**. FIG. **2** depicts a cross-sectional representation of mask **106** formed over lithium niobate layer **104**, according to some embodiments. While a single “island” of mask **106** is depicted in FIG. **2**, it is to be understood that mask **106** may include several islands or other structures defined in a pattern across the surface of lithium niobate layer **104**. Mask **106** may be, for example, a proton etch mask or other metal hard mask. Any material that is stable in hydrogen rich acid (such as benzoic acid) may be used for mask **106**. In one contemplated embodiment, mask **106** is formed by positioning a titanium layer on lithium niobate layer **104** and then defining a pattern for mask **106** using electron-beam lithography (e-beam lithography) and using liftoff. Other embodiments using photolithography and dry/wet etch may be contemplated for forming mask **106**.

(34) In various embodiments, mask **106** protects areas of lithium niobate layer **104** with dimensions larger than the dimensions intended for any final structures of lithium niobate. Larger areas may be protected by mask **106** to allow for over-etching of lithium niobate or other materials with the final dimensions determined by subsequent processing. After mask **106** is in position, the surface of substrate **100** may be exposed to a proton exchange treatment, as shown in FIG. **2**. Proton exchange

treatment of substrate **100** may include, for example, submerging the substrate in hydrogen rich acid (such as benzoic acid) at an elevated temperature for a period of time. For example, in one embodiment, substrate **100** may be submerged for 30 minutes in benzoic acid at a temperature of about 280° C. Other hydrogen rich acids may also be implemented such as, but not limited to, citric acid or acetic acid. In various embodiments, the proton exchange treatment is an isotropic process (e.g., non-directional process), as shown in FIG. 2

(35) FIG. 3 depicts a cross-sectional representation of substrate **100** after the proton exchange treatment, according to some embodiments. During the proton exchange treatment, hydrogen from the acid replaces lithium ions in unprotected (e.g., unmasked) portions of lithium niobate layer **104**. For instance, as shown in FIG. 3, unprotected portions of lithium niobate layer **104** become proton exchanged regions **108**. The depth of proton exchange in regions **108** may depend on time, temperature, and/or acid strength. The time and temperature for a specific thickness of lithium niobate layer **104** may be determined, for example, using a diffusion equation.

(36) After the proton exchange treatment, proton exchanged regions **108** may be removed using a wet etch process. FIG. 4 depicts a cross-sectional representation of substrate **100** after the wet etch process, according to some embodiments. In the illustrated embodiment, mask **106** has also been removed from substrate **100**. Mask **106** may be removed before, after, or during the wet etch process depending on the materials used for the mask and the wet etch. For instance, in embodiments with a titanium mask and hydrofluoric (HF) acid for wet etching, the titanium mask may be removed during the wet etch process.

(37) In one embodiment, the wet etch process is implemented by placing substrate **100** in concentrated hydrofluoric (HF) acid. HF acid is selective in removing proton exchanged regions **108** (e.g., protonated lithium niobate) versus non-proton exchanged regions (e.g., protected regions of lithium niobate layer **104** with unprotonated lithium niobate). For instance, in one example, HF acid may remove protonated lithium niobate at an etch rate of about 32 nm/min while unprotonated lithium niobate is removed at a rate of less than about 1 nm/min. Accordingly, an etch time may be implemented to minimize the removal of unprotonated lithium niobate. Additional embodiments may be contemplated that implement a dry etch process that is selective between protonated lithium niobate and unprotonated lithium niobate.

(38) After the wet etch process, unprotonated lithium niobate in lithium niobate layer **104** is left as lithium niobate structure **109**. Lithium niobate structure **109** may be an island structure or other three-dimensional structure on silicon substrate **102**. As is known in the semiconductor processing industry, silicon in silicon substrate **102** is largely unaffected by exposure to HF acid. Accordingly, a smooth, relatively unaffected silicon substrate **102** surface is left surrounding lithium niobate structure **109**.

(39) After lithium niobate structure **109** is defined on silicon substrate **102**, a mask defining a final (and more finely tuned) lithium niobate structure may be formed on substrate **100**. FIG. 5 depicts a cross-sectional representation of substrate **100** with lithium niobate structure mask **110**, according to some embodiments. In the illustrated embodiment, mask **110** is formed on lithium niobate structure **109**. In various embodiments, mask **110** is formed from silicon dioxide (SiO₂). For example, silicon dioxide may be spin-on silicon dioxide deposited on substrate **100** and then patterned using e-beam lithography.

(40) In certain embodiments, mask **110** defines boundaries and dimensions in lithium niobate structure **109** such that an acoustic resonator structure and anchors can be formed from the lithium niobate structure **109** during further processing. For example, in some embodiments, mask **110** may define minimum dimensions of about 100 nm. E-beam lithography may be suitable for defining such dimensions. Other patterning methods may include, but not be limited to, deep-UV photolithography, resist mask patterning, or other hard mask patterning.

(41) In various embodiments, a mask is also placed over exposed portions of silicon substrate **102**. FIG. 6 depicts a cross-sectional representation of substrate **100** with lithium niobate structure mask

110 and silicon substrate mask **112**, according to some embodiments. Mask **110** and mask **112** may be etch masks for protection of underlying features during further processing by an etch process. In some embodiments, mask **112** is formed in a separate, subsequent processing step from mask **110**, even in instances where mask **110** and mask **112** are formed from the same material. In some embodiments, mask **112** is formed at the same time and from the same materials as mask **110**. Mask **112** may protect large portions of silicon substrate **102** outside lithium niobate structure **109** from being damaged or affected during subsequent etch processing to form the final, more detailed lithium niobate structure.

(42) In the illustrated embodiment, mask **112** is spaced a small distance away from the walls of lithium niobate structure **109** (represented by gap **114** in FIG. 6). Gap **114** may be allowed during processing to avoid inadvertent covering of lithium niobate structure **109** by mask **112**. Various embodiments may, however, be contemplated where gap **114** is not present and mask **112** is close to the walls of lithium niobate structure **109**. In such embodiments, there may be overlap between mask **112** and the walls of lithium niobate structure **109**.

(43) After mask **110** and mask **112** are positioned on substrate **100**, an etch process may be implemented to form a lithium niobate structure with final dimensions (e.g., dimensions for an acoustic resonator structure). FIG. 7 depicts a cross-sectional representation of substrate **100** with lithium niobate structure mask **110** and silicon substrate mask **112** being subjected to a dry etch process, according to some embodiments. The dry etch process may be performed using gases or dry chemicals known to be capable of removing unprotonated lithium niobate. In certain embodiments, the dry etch process is an anisotropic (e.g., directional) dry etch process. In some embodiments, a brief HF acid dip or other mask removal process may be used after dry etching to clear masks **110** and **112** from substrate **100**.

(44) As shown in FIG. 7, gap **114** may allow some etching of silicon substrate **102** between walls of lithium niobate structure **109** and mask **112**. Such etching may form “moats” or other concave structures in silicon substrate **102**, as shown in FIG. 8 below. In embodiments where gap **114** is not present and mask **112** overlaps the walls of lithium niobate structure **109**, there may be some height variation across the exposed area of the lithium niobate structure. This outcome may, however, be minimal due to relaxed minimum dimension rules and alignment requirements during lithography and/or the use of thick resist during lithography.

(45) FIG. 8 depicts a cross-sectional representation of substrate **100** with lithium niobate structure **116** formed by the dry etch process and after mask removal, according to some embodiments. In the illustrated embodiment, lithium niobate structure **116** is formed as defined by the dimensions of mask **110** (shown in FIG. 7). Lithium niobate structure **116** may have the dimensions and structure defined for an acoustic resonator structure in a quantum computing device.

(46) As shown in FIG. 7, mask **112** protects various portions of silicon substrate **102** from the dry etch process. Accordingly, in FIG. 8, these protected portions form regions **118** that have relatively pristine silicon substrate. For instance, regions **118** have a surface smoothness (which may be defined by surface roughness, as described herein) on the order of a polished silicon substrate. Conversely, areas **120** along the walls of lithium niobate structure **116** have some etching/roughening of silicon substrate **102**. Additionally, in embodiments with gaps **114** during the dry etch process (shown in FIG. 7), the gaps may allow etching/roughening that forms concave structures **122** (e.g., moats) in silicon substrate **102**. However, as most of silicon substrate **102** has been protected by mask **112**, regions **118** are larger in scale than areas **120** and concave structures **122**.

(47) After the formation of lithium niobate structure **116** on silicon substrate **102**, further processing may be implemented to generate a quantum computing device from the silicon substrate. Further processing may include, for example, the deposition and patterning of superconducting and/or metal layers to form devices. FIG. 9 depicts a cross-sectional representation of substrate **100** with superconducting electrode layer **124** formed on the substrate,

according to some embodiments. In the illustrated embodiment, superconducting electrode layer **124** is formed over surfaces of both silicon substrate **102** and lithium niobate structure **116**. In certain embodiments, superconducting electrode layer **124** is formed using a directional (e.g., anisotropic) deposition process such that little to none of the superconducting electrode layer is deposited on vertical walls of lithium niobate structure **116**, as shown in FIG. **9**. In some embodiments, superconducting electrode layer **124** is formed using a conformal deposition process. Using a conformal deposition process may reduce the need for later deposition of a bandage layer (e.g., bandage layer **132**, shown in FIG. **11**).

(48) In some embodiments, opening(s) **126** in superconducting electrode layer **124** is formed (e.g., patterned) above silicon substrate **102** and opening **128** in superconducting electrode layer **124** is formed (e.g., patterned) above lithium niobate structure **116**. Openings **126** and **128** may allow contact to silicon substrate **102** and/or lithium niobate structure **116** for further structures formed on substrate **100**. In certain embodiments, openings **126** are formed over portions of silicon substrate **102** having a smooth surface (e.g., regions **118** described above). In various embodiments, the smooth surface of the portions of silicon substrate **102** under openings **126** may be defined by a surface roughness measurement that can be measured by metrology techniques (e.g., a measure for roughness may be a root mean square of the surface profile, expressed in nm). For instance, in certain embodiments, the portions of silicon substrate **102** under openings **126** may have a surface roughness of at most about 2 nm. In some embodiments, the portions of silicon substrate **102** under openings **126** may have a surface roughness of at most about 1 nm. The portions of silicon substrate **102** under openings **126** having a smooth surface may produce a higher quality device as the smooth surface is positioned under a nonlinear element (e.g., nonlinear element **130**, described herein). Such devices with smooth surfaces under the nonlinear element may, for example, have higher coherence times than devices with rough surface under the nonlinear element, as described below.

(49) In certain embodiments, superconducting electrode layer **124** is formed over a large area of substrate **100**. Superconducting electrode layer **124** may be, for example, an aluminum layer formed over a large area of substrate **100**. Superconducting electrode layer **124** may then be patterned to provide desired structures (e.g., openings **126**, **128**) on the layer. Patterning may be implemented, for example, using e-beam patterning along with evaporation and liftoff or a dry etch process. In various embodiments, superconducting electrode layer **124** is a base electrode layer that is patterned to form various electrodes that may be utilized in a quantum computing device. Examples of electrodes that may be patterned and implemented in superconducting electrode layer **124** include, but are not limited to, an electrode that actuates and reads out mechanical motion, a transmission line, a waveguide resonator, a ground plane, or combinations thereof.

(50) In certain embodiments, after superconducting electrode layer **124** is deposited and patterned on substrate **100**, additional processing is provided to form additional structures on the substrate. FIG. **10** depicts a cross-sectional representation of substrate **100** with nonlinear element **130** formed on the substrate, according to some embodiments. In the illustrated embodiment, nonlinear element **130** is formed at the location of opening **126** on silicon substrate **102** (e.g., above the smooth surface of the silicon substrate). In certain embodiments, nonlinear element is a nonlinear superconducting element. In various embodiments, nonlinear element **130** is a junction-based nonlinear superconducting element or a transmon. In one embodiment, nonlinear element **130** is a Josephson Junction-based transmon qubit. Nonlinear element **130** may, however, be any other nonlinear superconducting device for implementation in a quantum processor.

(51) One contemplated method for forming nonlinear element **130** includes using three-dimensional e-beam patterning with a multilayer e-beam resist followed by aluminum evaporation (e.g., double angle aluminum evaporation) with oxidation in between aluminum evaporations steps for insulation formation, and then liftoff. Such a method provides electrical access to two separate conducting aluminum layers in addition to a thin insulating (oxidation) layer between the

aluminum layers. In various embodiments, the three-dimensional e-beam patterning forms a Dolan-Bridge Junction in nonlinear element **130**.

(52) In various embodiments, there may be an uncertainty in connection along superconducting electrode layer **124** due to process variations. For example, if directional deposition of superconducting electrode layer **124** is implemented, vertical or steep walled structures may have little to no electrode layer formed on the walls, which may cause defects in the device. Thus, in certain embodiments, a bandage layer (e.g., a metal layer) may be formed on substrate **100** to ensure an uninterrupted superconducting path between elements on the substrate. For instance, a bandage layer may be formed to ensure an uninterrupted superconducting path between nonlinear element **130**, an acoustic resonator electrode at lithium niobate structure **116**, a ground plane, and other circuit elements.

(53) FIG. **11** depicts a cross-sectional representation of substrate **100** with bandage layer **132** formed on the substrate, according to some embodiments. Bandage layer **132** may be a metal (e.g., aluminum) layer that is patterned to form connections at possible break points in superconducting electrode layer **124**. For example, bandage layer **132** may be patterned to be positioned along and near walls of lithium niobate structure **116** and near nonlinear element **130**, as shown in FIG. **11**.

(54) In various embodiments, bandage layer **132** is formed by patterning using e-beam lithography. After e-beam lithography patterning, ion milling may be used to remove native oxide. Evaporation for aluminum deposition and liftoff is then implemented to form bandage layer **132** in desired locations. In some embodiments, photolithography may be used for patterning. In some embodiments, multiple resist layers and three-dimensional e-beam lithography may be implemented to create metal (aluminum) bridges in bandage layer **132**. Bridges may provide extra metallization layering to enable better ground plane connectivity and/or flexible routing.

(55) FIG. **12** depicts a cross-sectional representation of substrate **100** with acoustic resonator **140** and nonlinear element **130** formed on the substrate, according to some embodiments. In the illustrated embodiment, acoustic resonator **140** is formed by the combination of lithium niobate structure **116**, superconducting electrode layer **124**, and bandage **132** along with void **134** formed in silicon substrate **102**. Void **134** may be formed by forming a mask (e.g., a release mask) on substrate **100** that is patterned via lithography with a thick resist layer. Silicon in silicon substrate **102** under the mask is then etched (e.g., using a XeF₂ isotropic silicon etch) to form void **134**, thereby forming acoustic resonator **140**.

(56) Device **200**, shown in FIG. **12**, represents an example of a quantum computing device formed on the original substrate **100** (shown in FIG. **1**) by the combination of acoustic resonator **140** and nonlinear element **130**. In certain embodiments, as shown in FIG. **12**, acoustic resonator **140** is positioned in proximity to nonlinear element **130** on silicon substrate **102** with at least some spacing between the nonlinear element and the acoustic resonator on the silicon substrate. Accordingly, acoustic resonator **140** and nonlinear element **130** are spaced apart on silicon substrate **102** and in device **200**.

(57) In certain embodiments, because nonlinear element **130** is formed over a smooth portion of silicon substrate **102**, as described above, device **200** has properties on the order of baseline qubits fabricated on high resistivity pristine silicon substrate devices. Additionally, device **200** has better electrical properties than devices with rough silicon surfaces underneath the nonlinear element. For example, device **200** may have coherence times of at least about 10 μ s while devices with rough surfaces have coherence times on the order of 1 μ s. In some embodiments, device **200** may have higher coherence times. For example, in various embodiments, device **200** may have coherence times of at least about 90 μ s. Further refinement of processing may be further capable of increasing the coherence times.

(58) In embodiments with bridges in bandage layer **132**, formation of void **134** (shown in FIG. **12**) may be implemented before formation of the bandage layer. In such embodiments, fewer process steps with suspended structures may be implemented. The order of steps may be dependent on

which implementation is more stable (e.g., with or without bridges). Alternatively, bridges may be fabricated with separately patterned scaffolding if the scaffolding is amenable to dry or vapor release.

(59) In some contemplated embodiments, the steps described in FIGS. 5-8 (e.g., forming mask **110** and mask **112** followed by dry etching to form lithium niobate structure **116**) may be performed before the steps described in FIGS. 2-4 (e.g., forming mask **106**, proton exchange treatment, and protonated lithium niobate removal). FIGS. 13-20 depict cross-sectional representations of various steps in such a fabrication method embodiment. FIG. 13 depicts a cross-sectional representation of substrate **100** with mask **110** and mask **112** formed on lithium niobate layer **104**, according to some embodiments. FIG. 14 depicts a cross-sectional representation of substrate **100** from FIG. 13 being dry etched, according to some embodiments.

(60) FIG. 15 depicts a cross-sectional representation of substrate **100** from FIG. 13 after dry etching and mask removal, according to some embodiments. As shown in FIG. 15, forming mask **110** and mask **112** followed by dry etching to form lithium niobate structure **116** may leave the patterned lithium niobate structure **116** along with a large area of substrate **100** covered with lithium niobate regions **152** (e.g., unprotonated lithium niobate). Areas **150** may be positioned between lithium niobate structure **116** and regions **152**. In various embodiments, areas **150** may have a roughened silicon surface due to overetching from the dry etch process (similar to areas **120**, shown in FIG. 8).

(61) After the structure shown in FIG. 15 is formed, mask **106** (e.g., the proton exchange mask) may be formed to cover lithium niobate structure **116**. FIG. 16 depicts a cross-sectional representation of substrate **100** from FIG. 15 with mask **106** formed, according to some embodiments. In the illustrated embodiment, mask **106** is formed conformally over lithium niobate structure **116** to protect the walls of the lithium niobate structure. Proton exchange treatment is then performed on substrate **100**, as shown in FIG. 17. FIG. 18 depicts a cross-sectional representation of substrate **100** from FIG. 16 after proton exchange, according to some embodiments. As shown in FIG. 18, lithium niobate regions **152** are protonated to form proton exchanged regions **108** while lithium niobate structure **116** remains intact as unprotonated lithium niobate under mask **106**. Proton exchanged regions **108** and mask **106** may then be removed by HF acid etching to form the structure of substrate **100** shown in FIG. 19.

(62) Deposition of superconducting electrode layer **124** and formation of nonlinear element **130** may then proceed similarly to the steps depicted in FIGS. 9-12 to form device **200'** on substrate **100**, shown in FIG. 20. In the illustrated embodiment of FIG. 20, superconducting electrode layer **124** and bandage layer **132** are flatter (e.g., are less stepped) in device **200'** than for the embodiment of device **200** shown in FIG. 12. The embodiment of device **200'** in FIG. 20 may have similar electrical properties to the embodiment of device **200** in FIG. 12.

(63) In various embodiments, an additional wet etch HF acid (or other etchant) treatment may be implemented at any integration point during fabrication of device **200**. Such an additional etchant treatment may be beneficial as an acoustic resonator surface treatment (e.g., a treatment of a surface of lithium niobate structure **116**) since the additional etchant treatment may remove a minimal number of atomic layers. Removing one or more atomic layers can reduce surface TLS (two-level-system) density and improve intrinsic mechanical qualities in the device. In embodiments with additional etchant treatment, dimension changes due to the additional etching may be taken into account during other processing steps and/or the design of the device.

(64) In some contemplated embodiments, the deposition and patterning of superconducting electrode layer **124** may be performed on the structure of substrate **100** shown in FIG. 4 with the steps described in FIGS. 5-8 (e.g., forming mask **110** and mask **112** followed by dry etching to form lithium niobate structure **116**) being performed after the superconducting electrode layer **124** is formed. In such embodiments, the patterning of the acoustic resonator (e.g., patterning using mask **110** and mask **112**) is aligned with the ground plane formed by superconducting electrode

layer **124**. Mask **110** and mask **112** may be resist masks used for protection during dry etching. Alternatively, hard masks may be used if dry etching will not consume the hard masks and the hard masks can be removed without affecting superconducting material or lithium niobate.

(65) Example Fabrication Method of Standard Cell

(66) FIG. **21** is a flow diagram illustrating a method of fabrication of device **200**, according to some embodiments. At **2102**, in the illustrated embodiment, the method includes forming, from a layer of protonatable piezoelectric material positioned on a silicon substrate, a structure of the protonatable piezoelectric material on the silicon substrate where the structure of the protonatable piezoelectric material is formed by defining and removing portions of the layer of the protonatable piezoelectric material on the silicon substrate through a combination of: (a) application of a proton exchange treatment to the layer of protonatable piezoelectric material with a patterned mask to cause proton exchange in exposed portions of the protonatable piezoelectric material followed by wet etching the layer of protonatable piezoelectric material to remove proton exchanged portions of the protonatable piezoelectric material and (b) removing portions of the layer of protonatable piezoelectric material using a dry etch process.

(67) At **2104**, in the illustrated embodiment, the method includes forming an electrode layer on the silicon substrate where at least a portion of the electrode layer is formed over the structure of the protonatable piezoelectric material.

(68) At **2106**, in the illustrated embodiment, the method includes forming a nonlinear element on the electrode layer at a position spaced apart from the structure of the protonatable piezoelectric material.

(69) At **2108**, in the illustrated embodiment, the method includes removing at least some of the silicon substrate under the structure of protonatable piezoelectric material to form a void below the structure of protonatable piezoelectric material.

(70) Illustrative Computer System

(71) FIG. **22** is a block diagram illustrating an example computing device that may be used in at least some embodiments.

(72) FIG. **22** illustrates such a general-purpose computing device **2200** as may be used in any of the embodiments described herein. In the illustrated embodiment, computing device **2200** includes one or more processors **2210** coupled to a system memory **2220** (which may comprise both non-volatile and volatile memory modules) via an input/output (I/O) interface **2230**. Computing device **2200** further includes a network interface **2240** coupled to I/O interface **2230**.

(73) In various embodiments, computing device **2200** may be a uniprocessor system including one processor **2210**, or a multiprocessor system including several processors **2210** (e.g., two, four, eight, or another suitable number). Processors **2210** may be any suitable processors capable of executing instructions. For example, in various embodiments, processors **2210** may be general-purpose or embedded processors implementing any of a variety of instruction set architectures (ISAs), such as the x86, PowerPC, SPARC, or MIPS ISAs, or any other suitable ISA. In multiprocessor systems, each of processors **2210** may commonly, but not necessarily, implement the same ISA. In some implementations, graphics processing units (GPUs) may be used instead of, or in addition to, conventional processors.

(74) System memory **2220** may be configured to store instructions and data accessible by processor(s) **2210**. In at least some embodiments, the system memory **2220** may comprise both volatile and non-volatile portions; in other embodiments, only volatile memory may be used. In various embodiments, the volatile portion of system memory **2220** may be implemented using any suitable memory technology, such as static random access memory (SRAM), synchronous dynamic RAM or any other type of memory. For the non-volatile portion of system memory (which may comprise one or more NVDIMMs, for example), in some embodiments flash-based memory devices, including NAND-flash devices, may be used. In at least some embodiments, the non-volatile portion of the system memory may include a power source, such as a supercapacitor or

other power storage device (e.g., a battery). In various embodiments, memristor based resistive random access memory (ReRAM), three-dimensional NAND technologies, Ferroelectric RAM, magnetoresistive RAM (MRAM), or any of various types of phase change memory (PCM) may be used at least for the non-volatile portion of system memory. In the illustrated embodiment, program instructions and data implementing one or more desired functions, such as those methods, techniques, and data described above, are shown stored within system memory **2220** as code **2225** and data **2226**.

(75) In some embodiments, I/O interface **2230** may be configured to coordinate I/O traffic between processor **2210**, system memory **2220**, and any peripheral devices in the device, including network interface **2240** or other peripheral interfaces such as various types of persistent and/or volatile storage devices. In some embodiments, I/O interface **2230** may perform any necessary protocol, timing or other data transformations to convert data signals from one component (e.g., system memory **2220**) into a format suitable for use by another component (e.g., processor **2210**). In some embodiments, I/O interface **2230** may include support for devices attached through various types of peripheral buses, such as a variant of the Peripheral Component Interconnect (PCI) bus standard or the Universal Serial Bus (USB) standard, for example. In some embodiments, the function of I/O interface **2230** may be split into two or more separate components, such as a north bridge and a south bridge, for example. Also, in some embodiments some or all of the functionality of I/O interface **2230**, such as an interface to system memory **2220**, may be incorporated directly into processor **2210**.

(76) Network interface **2240** may be configured to allow data to be exchanged between computing device **2200** and other devices **2260** attached to a network or networks **2250**, such as other computer systems or devices. In various embodiments, network interface **2240** may support communication via any suitable wired or wireless general data networks, such as types of Ethernet network, for example. Additionally, network interface **2240** may support communication via telecommunications/telephony networks such as analog voice networks or digital fiber communications networks, via storage area networks such as Fibre Channel SANs, or via any other suitable type of network and/or protocol.

(77) In some embodiments, system memory **2220** may represent one embodiment of a computer-accessible medium configured to store at least a subset of program instructions and data used for implementing the methods and devices discussed in the context of FIG. 1 through FIG. 18. However, in other embodiments, program instructions and/or data may be received, sent or stored upon different types of computer-accessible media. Generally speaking, a computer-accessible medium may include non-transitory storage media or memory media such as magnetic or optical media, e.g., disk or DVD/CD coupled to computing device **2200** via I/O interface **2230**. A non-transitory computer-accessible storage medium may also include any volatile or non-volatile media such as RAM (e.g., SDRAM, DDR SDRAM, RDRAM, SRAM, etc.), ROM, etc., that may be included in some embodiments of computing device **2200** as system memory **2220** or another type of memory. In some embodiments, a plurality of non-transitory computer-readable storage media may collectively store program instructions that when executed on or across one or more processors implement at least a subset of the methods and techniques described above. A computer-accessible medium may further include transmission media or signals such as electrical, electromagnetic, or digital signals, conveyed via a communication medium such as a network and/or a wireless link, such as may be implemented via network interface **2240**. Portions or all of multiple computing devices such as that illustrated in FIG. 22 may be used to implement the described functionality in various embodiments; for example, software components running on a variety of different devices and servers may collaborate to provide the functionality. In some embodiments, portions of the described functionality may be implemented using storage devices, network devices, or special-purpose computer systems, in addition to or instead of being implemented using general-purpose computer systems. The term “computing device”, as used

herein, refers to at least all these types of devices, and is not limited to these types of devices.

CONCLUSION

(78) Various embodiments may further include receiving, sending or storing instructions and/or data implemented in accordance with the foregoing description upon a computer-accessible medium. Generally speaking, a computer-accessible medium may include storage media or memory media such as magnetic or optical media, e.g., disk or DVD/CD-ROM, volatile or non-volatile media such as RAM (e.g., SDRAM, DDR, RDRAM, SRAM, etc.), ROM, etc., as well as transmission media or signals such as electrical, electromagnetic, or digital signals, conveyed via a communication medium such as network and/or a wireless link.

(79) The various methods as illustrated in the Figures and described herein represent exemplary embodiments of methods. The methods may be implemented in software, hardware, or a combination thereof. The order of method may be changed, and various elements may be added, reordered, combined, omitted, modified, etc.

(80) Various modifications and changes may be made as would be obvious to a person skilled in the art having the benefit of this disclosure. It is intended to embrace all such modifications and changes and, accordingly, the above description to be regarded in an illustrative rather than a restrictive sense.

Claims

1. A method, comprising: forming, from a layer of protonatable piezoelectric material positioned on a silicon substrate, a structure of protonatable piezoelectric material on the silicon substrate, wherein the structure of the protonatable piezoelectric material is formed by defining and removing portions of the layer of the protonatable piezoelectric material on the silicon substrate through a combination of: (a) application of a proton exchange treatment to the layer of protonatable piezoelectric material with a patterned mask to cause proton exchange in exposed portions of the protonatable piezoelectric material followed by wet etching the layer of protonatable piezoelectric material to remove proton exchanged portions of the protonatable piezoelectric material; and (b) removing portions of the layer of protonatable piezoelectric material using a dry etch process; forming an electrode layer on the silicon substrate; forming a nonlinear element on the electrode layer at a position spaced apart from the structure of the protonatable piezoelectric material; and removing at least some of the silicon substrate under the structure of protonatable piezoelectric material to form a void below the structure of protonatable piezoelectric material.
2. The method of claim 1, wherein the protonatable piezoelectric material is lithium niobate.
3. The method of claim 1, wherein removing portions of the layer of protonatable piezoelectric material using the dry etch process includes forming a dry etch mask defining a pattern on the layer of protonatable piezoelectric material and removing portions of the layer of protonatable piezoelectric material exposed by the dry etch mask.
4. The method of claim 1, wherein the structure of the protonatable piezoelectric material is formed by performing (a) then (b).
5. The method of claim 1, wherein the structure of the protonatable piezoelectric material is formed by performing (b) then (a).
6. The method of claim 1, wherein at least a portion of the electrode layer is formed over the structure of the protonatable piezoelectric material, the method further comprising forming a first opening through the electrode layer to the structure of the protonatable piezoelectric material and a second opening through the electrode layer to the silicon substrate.
7. The method of claim 1, wherein the nonlinear element is a nonlinear superconducting element, and wherein the electrode layer is a superconducting electrode layer.
8. The method of claim 1, wherein the nonlinear element is formed at a location of an opening in the electrode layer exposing the silicon substrate.

9. The method of claim 8, wherein the opening in the electrode layer is located over a smooth portion of the silicon substrate, the smooth portion of the silicon substrate having a surface smoothness on the order of a polished silicon substrate.
10. The method of claim 1, further comprising forming one or more of the following electrodes in the electrode layer: an electrode that actuates and reads out mechanical motion, a transmission line, a waveguide resonator, and a ground plane.
11. The method of claim 1, wherein removing the at least some of the silicon substrate under the structure of protonatable piezoelectric material includes: forming a release mask over a portion of the structure of the protonatable piezoelectric material and the electrode layer, leaving a portion of the silicon substrate exposed; and removing silicon under the exposed portion of the structure of the protonatable piezoelectric material using an isotropic silicon etch process.
12. The method of claim 1, further comprising forming a bandage metal layer on the silicon substrate, wherein at least a portion of the bandage metal layer contacts at least the portion of the electrode layer formed over the structure of the protonatable piezoelectric material, and wherein at least a portion of the bandage metal layer contacts a portion of the nonlinear element.
13. The method of claim 1, wherein the proton exchange treatment includes placing the silicon substrate in a hydrogen rich acid.
14. A non-transitory computer readable storage medium storing a plurality of instructions which, when executed, generate a device according to steps, comprising: forming, from a layer of protonatable piezoelectric material positioned on a silicon substrate, a structure of protonatable piezoelectric material on the silicon substrate, wherein the structure of the protonatable piezoelectric material is formed by defining and removing portions of the layer of the protonatable piezoelectric material on the silicon substrate through a combination of: (a) application of a proton exchange treatment to the layer of protonatable piezoelectric material with a patterned mask to cause proton exchange in exposed portions of the protonatable piezoelectric material followed by wet etching the layer of protonatable piezoelectric material to remove proton exchanged portions of the protonatable piezoelectric material; and (b) removing portions of the layer of protonatable piezoelectric material using a dry etch process; forming an electrode layer on the silicon substrate; forming a nonlinear element on the electrode layer at a position spaced apart from the structure of the protonatable piezoelectric material; and removing at least some of the silicon substrate under the structure of protonatable piezoelectric material to form a void below the structure of protonatable piezoelectric material.
15. The non-transitory computer readable storage medium of claim 14, wherein removing portions of the layer of protonatable piezoelectric material using the dry etch process includes forming a dry etch mask defining a pattern on the layer of protonatable piezoelectric material and removing portions of the layer of protonatable piezoelectric material exposed by the dry etch mask.
16. The non-transitory computer readable storage medium of claim 14, wherein the structure of the protonatable piezoelectric material is formed by performing (a) then (b).
17. The non-transitory computer readable storage medium of claim 14, wherein the structure of the protonatable piezoelectric material is formed by performing (b) then (a).
18. The non-transitory computer readable storage medium of claim 14, wherein at least a portion of the electrode layer is formed over the structure of the protonatable piezoelectric material, the steps further comprising forming a first opening through the electrode layer to the structure of the protonatable piezoelectric material and a second opening through the electrode layer to the silicon substrate.
19. The non-transitory computer readable storage medium of claim 14, the steps further comprising forming one or more of the following electrodes in the electrode layer: an electrode that actuates and reads out mechanical motion, a transmission line, a waveguide resonator, and a ground plane.
20. The non-transitory computer readable storage medium of claim 14, wherein removing the at least some of the silicon substrate under the structure of protonatable piezoelectric material

includes: forming a release mask over a portion of the structure of the protonatable piezoelectric material and the electrode layer, leaving a portion of the silicon substrate exposed; and removing silicon under the exposed portion of the structure of the protonatable piezoelectric material using an isotropic silicon etch process.
